

POLITECNICO
MILANO 1863

Uncovering latent themes and emotional dynamics youth precariousness

MSc Thesis Presentation

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Context

- Research activity with the Universitat Autonòma de Barcelona
- VulnYouth project (IGOP, Institut de Govern i Polítiques Públiques, UAB Barcelona, 2023)
- Funded by the La Caixa Foundation, Spain:
- aims to explore the link between mental health and job insecurity among young people in Spain.
- Question: *"What does precariousness mean to you?"*

UAB
Universitat Autònoma de Barcelona

"la Caixa" Foundation

Goal

Do an emotion Detection over the 2459 responses of the open-ended question.

After an initial preprocessing we got n = 2459 respondents and p = 68 covariates. We base our analysis on a set of variables including:

- Personal background;
- Working situation;
- Economic and Financial situation;
- Education;
- Household information;
- Open-ended question "What is precariousness for you?"

Examples of Open-Ended Responses

The following responses were provided by participants when asked to define "precariousness" in their own words:

- **Example 1: Living Standards**

  "Es no poder vivir tranquilo económica y socialmente. Es tener que vivir para trabajar y aun así no llegar a fin de mes."
  (It is not being able to live peacefully economically and socially. It is having to live to work and still not make ends meet.)

- **Example 2: Job Insecurity**

  "No tener un trabajo estable y que te paguen poco."
  (Not having a steady job and being paid little.)

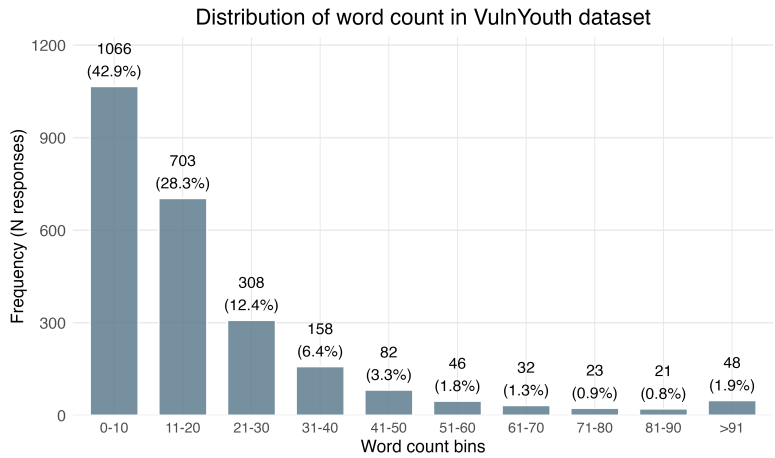
- **Example 3: Systemic Vulnerability**

  "No tener el poder de decidir, de negociar, de exigir una vida digna. Sentirse indigno e inferior."
  (Not having the power to decide, to negotiate, to demand a dignified life. Feeling unworthy and inferior.)

Working with open-ended questions

Context

- Traditional sociological metrics often fail to capture the **subjective experience** of precariousness.
- The *VulnYouth* study utilizes open-ended responses to allow Spanish youth to define their own conditions beyond rigid job insecurity indicators.



Problem

- The responses consist of very short texts, averaging 19.25 words per response with a standard deviation of 22.06.
- The corpus is almost exclusively focused on negative sentiment, making traditional lexicon-based approaches ineffective.

Solutions

1. Dimensional Emotion Detection (VAD)
2. Categorical Emotion Detection (RoBERTa)
3. Structural Topic Modeling (STM)

Detection emotion models using BERT

- Bidirectional Encoder Representations from Transformers (BERT) is a deep learning model based on the Transformer architecture (Devlin et al, 2019)
- It was build for classification of text
- it has around 110 Millions of parameters and was pretrained using a 3.3 billion words using a collection of 11,000 unpublished books and text extracted from Wikipedia, excluding lists, tables, and headers
- popular for its flexibility achived through so-called transfer learning:
- instead of starting from scratch, it uses parameters obtained during a pre-training phase: An output layer is thenadded to adapt the new model to specialize in a specific task.
- in our study, we use also RoBERTa, that is a bigger model than BERT (125 Millions of parameters)

- BERT was fine-tuned on EmoBank dataset, which has 3 continuous dimension emotions, all ranging from 1 to 5:
  - Valence: The degree of pleasantness or unpleasantness (Positive vs. Negative).
  - Arousal: The physiological and psychological intensity, ranging from high activation (e.g., anger) to low activation (e.g., calm or resignation)
  - Dominance: The perceived degree of control or agency the speaker feels over the situation.

Categorical emotion classification

- RoBERTa was fine-tuned on GoEmotions dataset, which has 27 distinct emotional labels: *admiration*, *amusement*, *approval*, *caring*, *curiosity*, *desire*, *excitement*, *gratitude*, *joy*, *love*, *optimism*, *pride*, *realization*, *relief*, *surprise*, *anger*, *annoyance*, *confusion*, *disappointment*, *disapproval*, *disgust*, *embarrassment*, *fear*, *grief*, *nervousness*, *remorse*, *sadness*
- The final output is a probability vector across all 28 classes,
- We selected the specific probability cut-off that maximized the F1-score for each of the 28 emotions independently in the validation set, and applied them to the test set.

- Structural Topic Modeling (STM) (Roberts et al., 2014) is a model that enables researchers to uncover topics directly from the data;
- STM is a mixed-membership model;
- A document is represented as a mixture of topics, with each word in the document assigned to a single topic;
- Each document can be expressed as a vector of probabilities θ_i, where $\theta_{i,k}$ is the probability that topic k is important to describe document i;
- aggiungi qualcosa che abbiamo usato la topic prevalence

General distribution of affect

- Valence: The mean valence score is 2.56 (SD=0.35). Precariousness is conceptualized almost exclusively through negative sentiment.
- Arousal: The mean arousal score is 2.95 (SD0.32), placing the discourse very close to the neutral threshold.
- Dominance: the mean dominance score is 2.73 (SD 0.28). This distribution is shifted towards the lower end of the scale of Dominance, indicating that respondents perceive precariousness as a state of low agency, controlled by external forces rather than their own choices (see Mohammad (2018)).

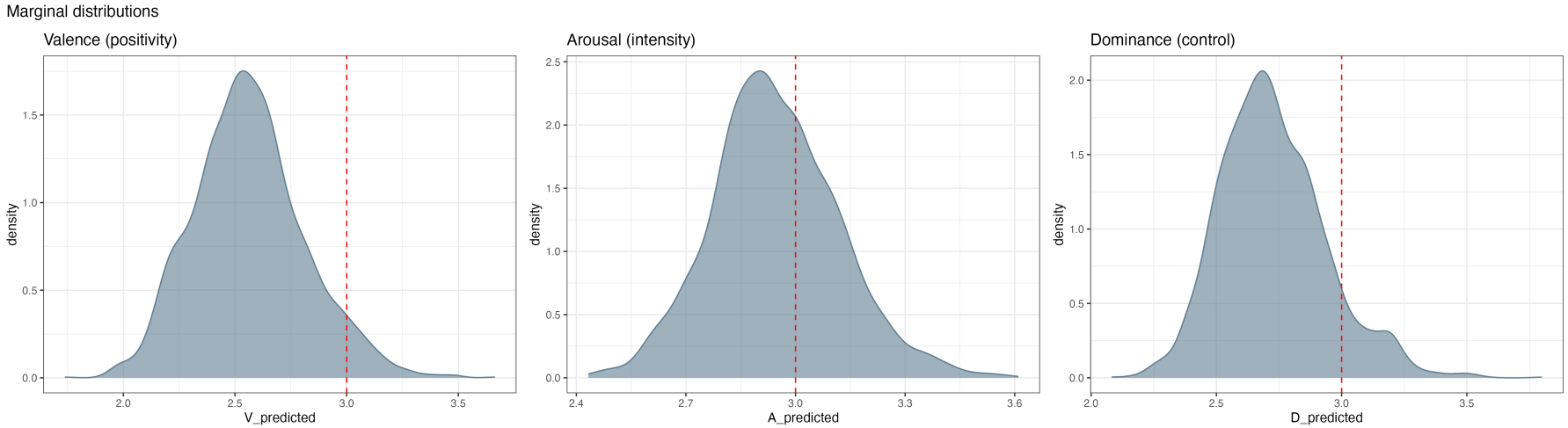


Figure: Marginal distributions of emotional dimensions.

Scatterplots

- Valence vs. Dominance: A strong positive correlation is visible, where lower valence scores consistently align with lower dominance scores.

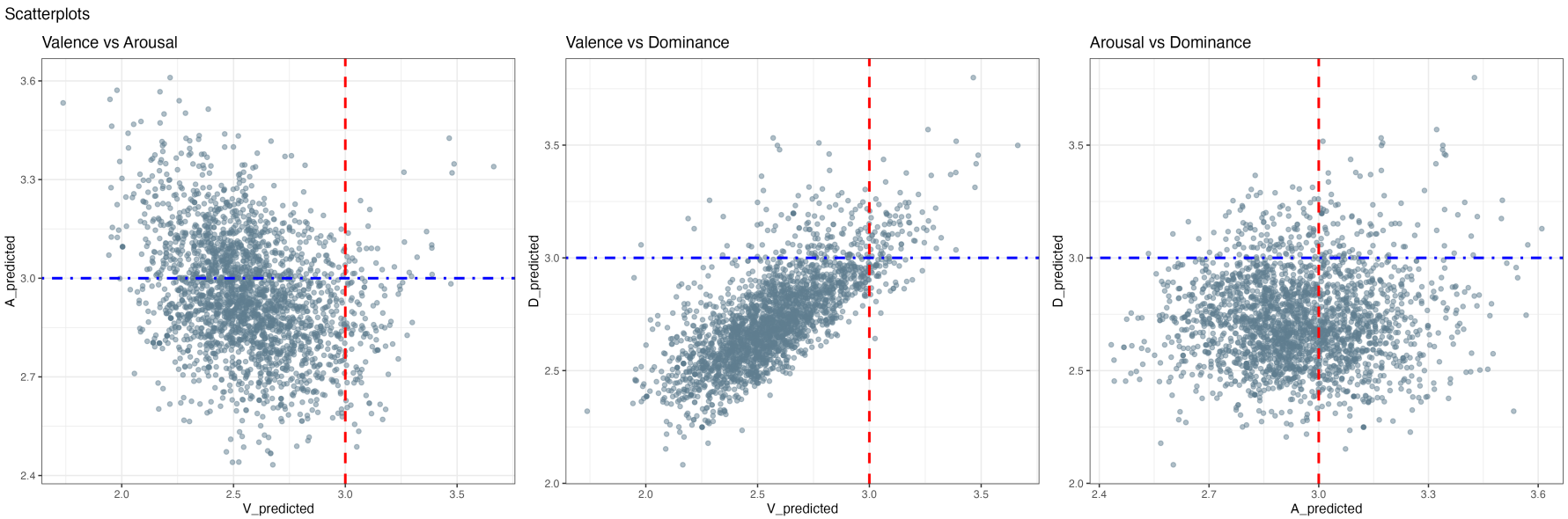


Figure: Pairwise scatterplots of the predicted emotional dimensions

Using the GoEmotions dataset, we find out that:

- Disappointment (51.2%): This is the most prevalent emotion, appearing in over half of the responses. It is described as "sadness or displeasure caused by the nonfulfillment of one's hopes or expectations"
- Sadness (48.8%); genericemo- tional pain or sorrow expressed by respondents.
- Disapproval (28.8%): many young people are actively judging the current socio-economic system as inadequate or wrong.
- Absence of high-arousal Negativity: Consistent with the dimensional VAD findings, high-arousal negative emotions are remarkably rare. Active states such as Anger (0.5%) and Fear (1.6%) appear with negligible frequency.

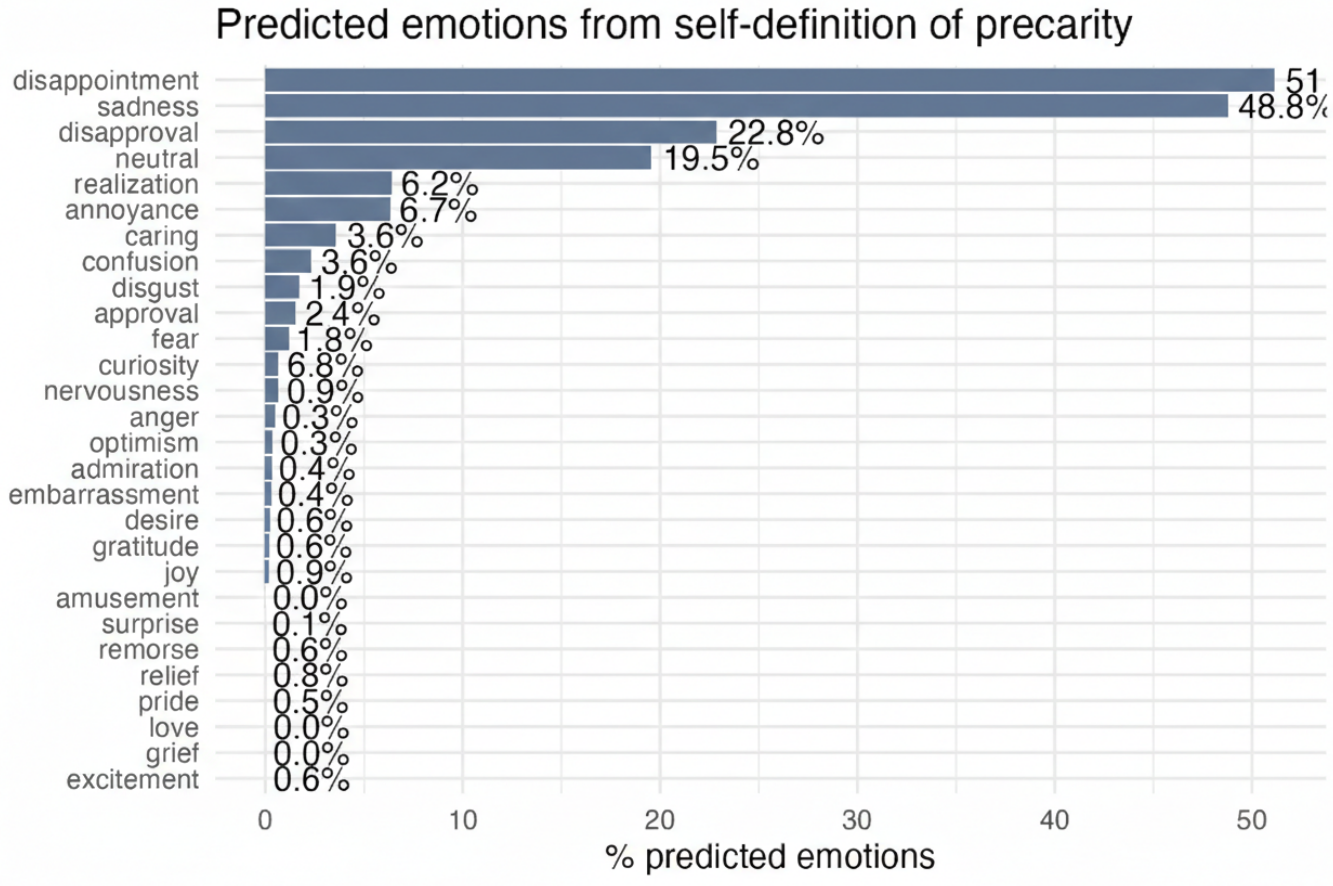
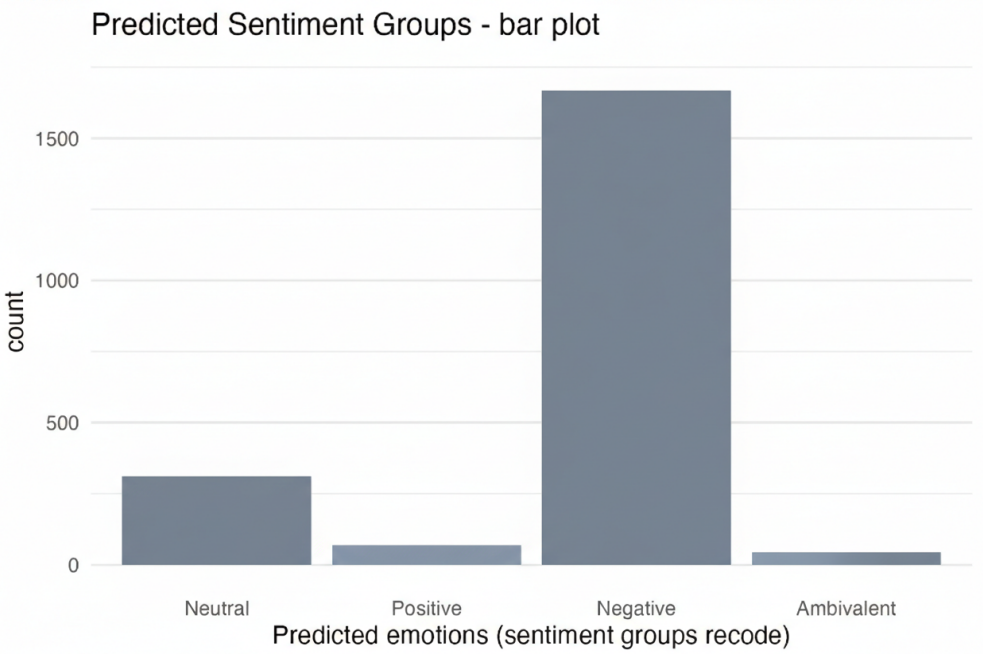


Figure: Frequency of predicted emotions in the open-ended questions of precariousness in the VulnYouth survey

Dividing emotions in 4 groups

- The 28 categorical emotional labels were aggregated into four high-level sentiment groups. Clearly Negative group dominates the scene.



The neutrality paradox

- we estimated two binary logistic regression models. Respondents were coded as emotional (1 = as- signed to Positive, Negative, or Ambivalent sen- timent groups) versus neutral (0 = exclusively Neutral category)
- FORMULA
- respondents experiencing high job insecurity have approximately 25% lower odds of being classified as emotional compared to those with stable employment.

Topic analysis: labels

- Topic 1: Resource-driven economic insecurity;
- Topic 2: Unaffordable independent living;
- Topic 3: Poor working conditions;
- Topic 4: Worker exploitation and vulnerability;
- Topic 5: Overtime without proper compensation;
- Topic 6: Illegal and unsafe working conditions;
- Topic 7: Not being valued in the workplace;
- Topic 8: No access to housing or food;
- Topic 9: Inability to meet living standards;
- Topic 10: Impossibility to save money;
- Topic 11: Living an unstable life;
- Topic 12: Impossibility to plan a future;
- Topic 13: Struggle to make ends meet;
- Topic 14: Lack of resources;
- Topic 15: Exploitation and Abusive Treatment;
- Topic 16: Not being able to afford basic needs;
- Topic 17: Impossibility of reaching economic independence;
- Topic 18: Low wages;
- Topic 19: Youth unemployment and insecurity;
- Topic 20: Job insecurity and insufficient stability;

Topic analysis and emotional predictors

We examine through a logistic regression whether specific predicted emotions are systematically associated with the probability of discussing particular topics.

Emotion	Predicting Topics	Notes
Sadness	T9, T13	Strong association.
Disappointment	T6, T13, T14	Statistically reliable, but thematically diffuse.
Disapproval	T3	Moderate effect size / AUC.
Neutral	T5, T18	Moderate effect size / AUC.
Fear	T1, T2, T3	**Strongest effect size**, but small sample proportion.

- **Moderate strength:** Associations for Disappointment, Disapproval, and Neutral are reliable but diffuse.
- **High Intensity:** *Fear* shows the highest concentration around structural conditions (T1, T2, T3).

This thesis applied NLP techniques (STM and Transformers) to the *VulnYouth* study to map the subjective experience of precariousness among Spanish youth.

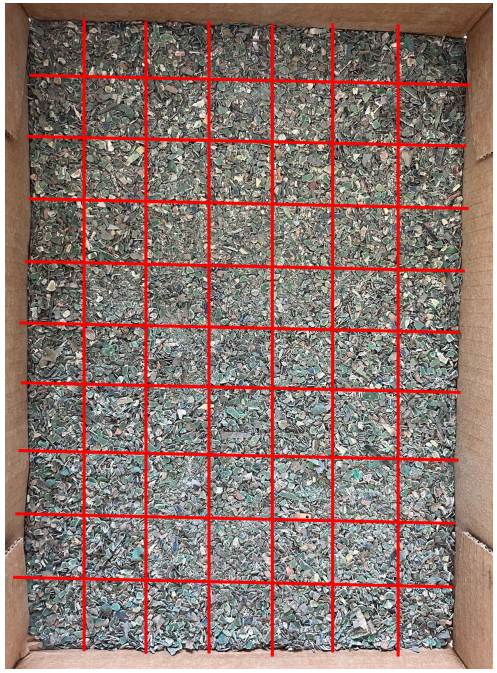
- **Emotional Signature:** Precariousness is defined by **Disappointment (51.2%)** and **Sadness (48.8%)**, rather than Anger or Fear.
- **The Neutrality Paradox:** High objective job insecurity leads to **neutral, detached language** rather than emotional descriptors.
- **Multidimensionality:** STM identified 20 latent themes, showing that precariousness spans beyond just wages to structural instability.
- **Methodological Impact:** Transformer architectures (BERT/RoBERTa) combined with STM effectively handle short, sparse sociological survey texts where lexicon-based methods fail.

References

Sampling design

The grinded PCB material is divided into a grid of 93 cells. Every cell is assigned the same reference mass

$$n = 10^6 \text{ parts} \approx 1.76 \text{ g}$$



Observed data

XRF measurements return elemental concentrations in **ppm**. For each sampled cell $i = 1, \ldots, I$ we observe

$$\boldsymbol{N}_i = (N_{i,1}, \ldots, N_{i,K}), \quad K = 40$$

with total mass $\sum_k N_{i,k} = 10^6$.

Latent composition

All samples share a common composition $\boldsymbol{p} \in \mathcal{S}^{K-1}$ representing the true elemental composition of the PCBs batch.

The statistical model is based on a multinomial sampling mechanism for the elemental composition vector

$$\boldsymbol{N}_i \mid \boldsymbol{p} \sim \text{Multinomial}(n, \boldsymbol{p}), \qquad \boldsymbol{p} \sim \pi_p,$$

where

$$\boldsymbol{p} = (p_1, \ldots, p_K) \in \mathcal{S}^{K-1}$$

represents the latent elemental proportions of the batch.

Different Bayesian models arise from different choices of the prior distribution π_p on the simplex.

As a baseline model we consider the **Dirichlet prior**, leading to the classical **Dirichlet–Multinomial model** (DM), which provides a simple reference for more flexible extensions.

Hierarchical model:

$$\mathbf{N}_i \mid \boldsymbol{p} \overset{\text{i.i.d.}}{\sim} \text{Multinomial}(n, \boldsymbol{p}). \quad i = 1, \ldots, I,$$
$$\boldsymbol{p} \sim \text{Dirichlet}(\alpha_1, \ldots, \alpha_K), \quad \boldsymbol{p} \in \mathcal{S}^{K-1}$$

Likelihood:

$$\mathcal{L}(\{\mathbf{N}_i\}, \boldsymbol{p} \mid \boldsymbol{\alpha}) = \prod_{i=1}^{I} \left[\underbrace{\frac{n!}{\prod_k N_{i,k}!} \prod_{k=1}^{K} p_k^{N_{i,k}}}_{\text{Multinomial}(\mathbf{N}_i \mid n, \boldsymbol{p})} \cdot \underbrace{\frac{1}{B(\boldsymbol{\alpha})} \prod_{k=1}^{K} p_k^{\alpha_k - 1}}_{\text{Dirichlet}(\boldsymbol{p}_i \mid \boldsymbol{\alpha})} \right],$$

Posteriors:

$$\boldsymbol{\alpha}_{post} \mid \mathbf{N}_i = \boldsymbol{\alpha}_{prior} + \sum_{i=0}^{I} \mathbf{N}_i$$
$$\boldsymbol{p} \mid \mathbf{N}_i, \boldsymbol{\alpha} \sim \text{Dirichlet}(\boldsymbol{\alpha}_{post}),$$

The **Flexible Dirichlet (FD)** distribution [**?**] extends the classical Dirichlet model by introducing additional flexibility in the dependence structure. Let the latent variables be defined as

$$Y_k = W_k + Z_k U, \qquad k = 1, \dots, K$$

where

$$W_k \sim \text{Gamma}(\alpha_k, 1), \quad U \sim \text{Gamma}(\tau, 1), \quad \boldsymbol{Z} \sim \text{Multinomial}(1, \boldsymbol{q})$$

The compositional vector is obtained by normalization

$$\boldsymbol{p} = \frac{\boldsymbol{Y}}{\sum_{k=1}^{K} Y_k}$$

and we write

$$\boldsymbol{p} \sim FD_K(\boldsymbol{\alpha}, \boldsymbol{q}, \tau). \tag{1}$$

A key property of the Flexible Dirichlet distribution is its **finite mixture representation**

$$f_{FD}(\boldsymbol{p}) = \sum_{j=1}^{K} q_j \, \text{Dir}(\boldsymbol{p} \mid \boldsymbol{\alpha} + \tau \boldsymbol{e}_j). \tag{2}$$

This leads to the following hierarchical model for the ppm data.

$$\begin{aligned} \boldsymbol{N}_i \mid \boldsymbol{p} &\sim \text{Multinomial}(n, p_1, \ldots, p_K) \\ \boldsymbol{p} \mid \boldsymbol{z} = \boldsymbol{e}_j &\sim \text{Dirichlet}(\boldsymbol{\alpha} + \tau \boldsymbol{e}_j) \\ P(\boldsymbol{z} = \boldsymbol{e}_j) &= q_j, \\ \boldsymbol{q} &\sim \text{Dirichlet}(\boldsymbol{\beta}) \end{aligned}$$

$$\pi(\boldsymbol{\alpha}, \tau) \propto b^{\tau} \prod_{k=1}^{K} a_i^{\alpha_i}$$

The FD prior allows a richer dependence structure than the Dirichlet model and can exhibit multimodality, corresponding to multiple regimes.

Let $\alpha_+ = \sum_{k=1}^{K} \alpha_k$. The first moment of the FD distribution is

$$\mathbb{E}(p_k) = \frac{\alpha_k + q_k \tau}{\alpha_+ + \tau}.$$

For $k \neq r$, the covariance between components is

$$\mathrm{Cov}(p_k, p_r) = -\frac{\mathbb{E}(p_k)\mathbb{E}(p_r)}{\alpha_+ + \tau + 1} - \frac{\tau^2 q_k q_r}{(\alpha_+ + \tau)(\alpha_+ + \tau + 1)}. \tag{3}$$

Compared with the Dirichlet distribution, the FD model allows **stronger dependence between components**, while covariances remain negative due to the simplex constraint.

Given the hierarchical model

$$\begin{aligned} \boldsymbol{N}_i \mid \boldsymbol{p} &\sim \text{Multinomial}(n, \boldsymbol{p}) \\ \boldsymbol{p} \mid \boldsymbol{z} = \boldsymbol{e}_j &\sim \text{Dirichlet}(\boldsymbol{\alpha} + \tau \boldsymbol{e}_j) \end{aligned}$$

posterior inference can be performed through Gibbs sampling.

The main full conditional distributions are

$$\boldsymbol{p} \mid \boldsymbol{z}, \boldsymbol{\alpha}, \tau, \{\boldsymbol{N}_i\} \sim \text{Dirichlet}\left(\boldsymbol{\alpha} + \tau \boldsymbol{e}_j + \sum_{i=1}^{I} \boldsymbol{N}_i\right)$$

$$P(\boldsymbol{z} = \boldsymbol{e}_j \mid \boldsymbol{p}, \boldsymbol{q}, \boldsymbol{\alpha}, \tau) = \frac{q_j \, \text{Dir}(\boldsymbol{p} \mid \boldsymbol{\alpha} + \tau \boldsymbol{e}_j)}{\sum_{k=1}^{K} q_k \, \text{Dir}(\boldsymbol{p} \mid \boldsymbol{\alpha} + \tau \boldsymbol{e}_k)}$$

$$\boldsymbol{q} \mid \boldsymbol{z} = \boldsymbol{e}_j \sim \text{Dirichlet}(\beta_1, \ldots, \beta_j + 1, \ldots, \beta_K).$$

The remaining parameters, namely $\boldsymbol{\alpha}$ and τ, are associated with more complex posterior distributions, for which closed–form expressions are not available.